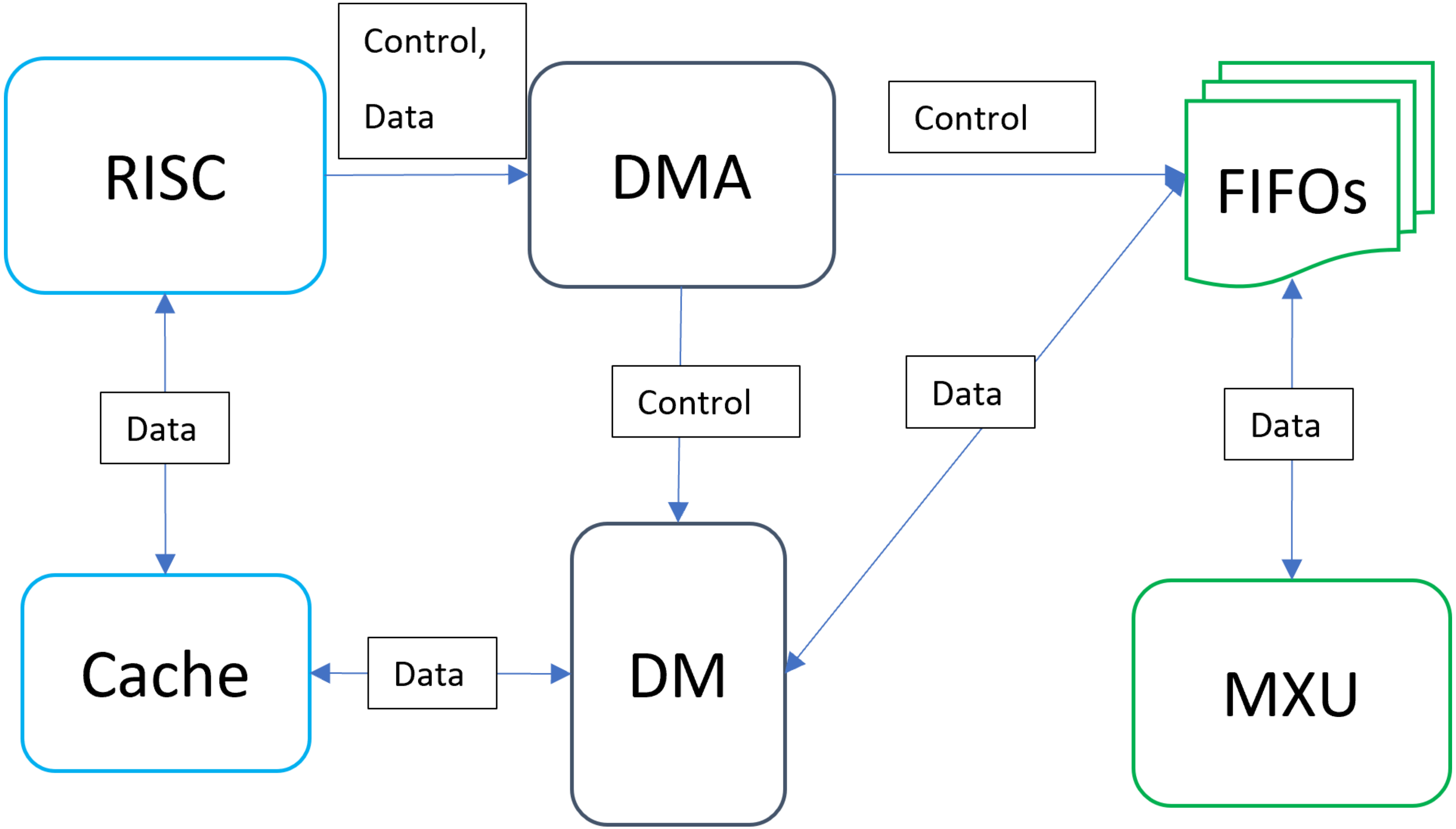
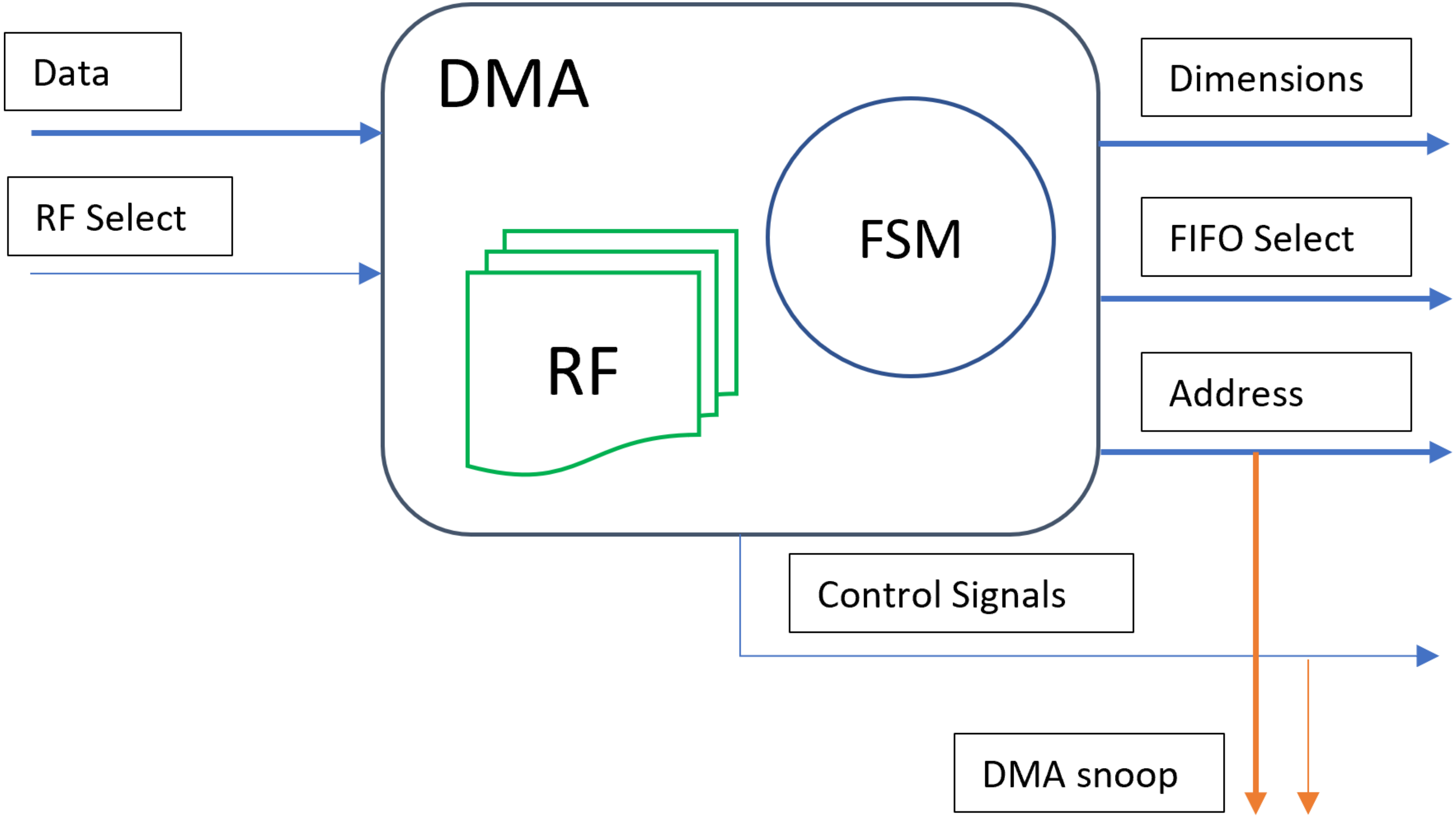
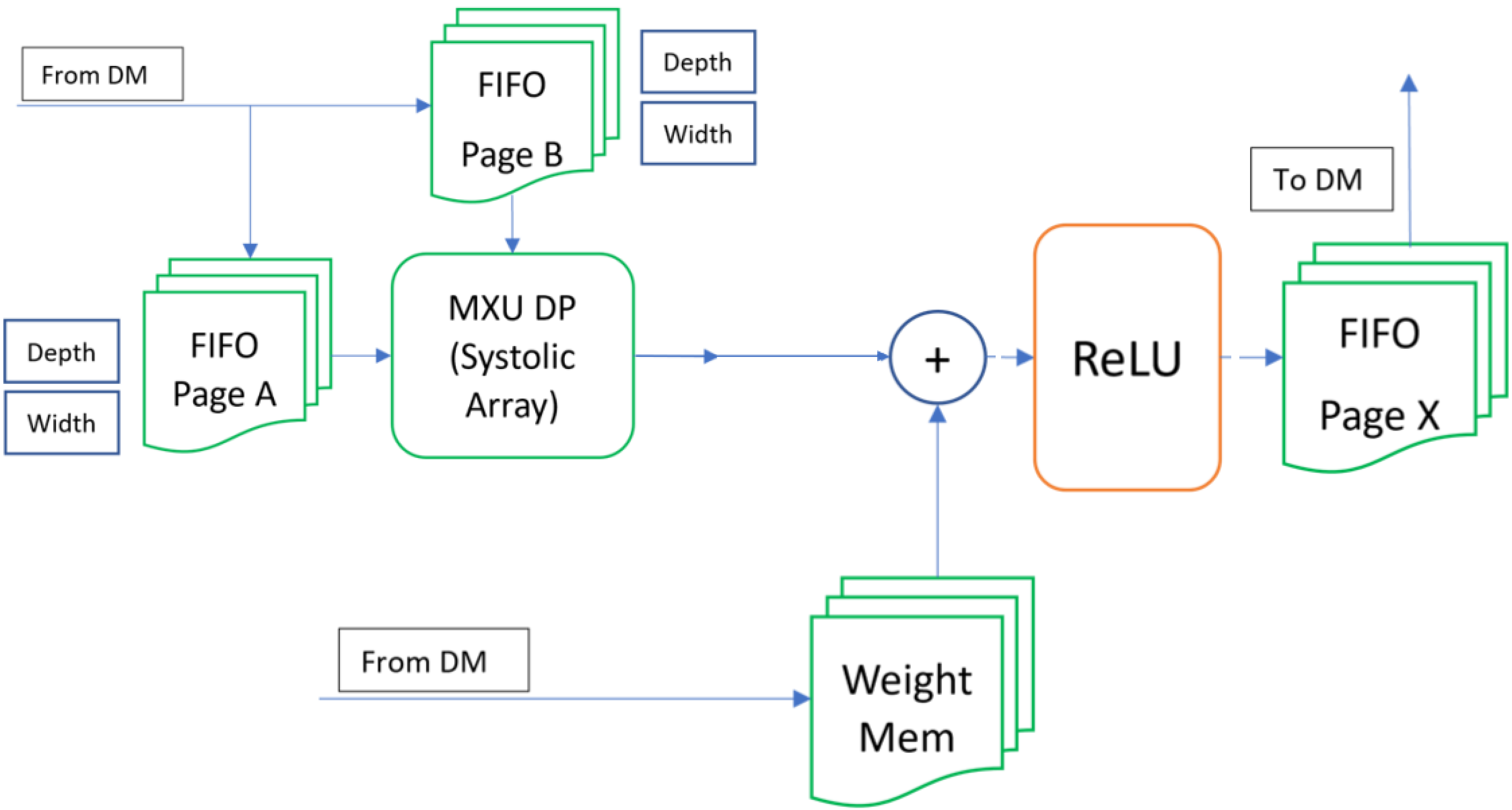
Heterogeneous AI Accelerator for Feedforward Calculations

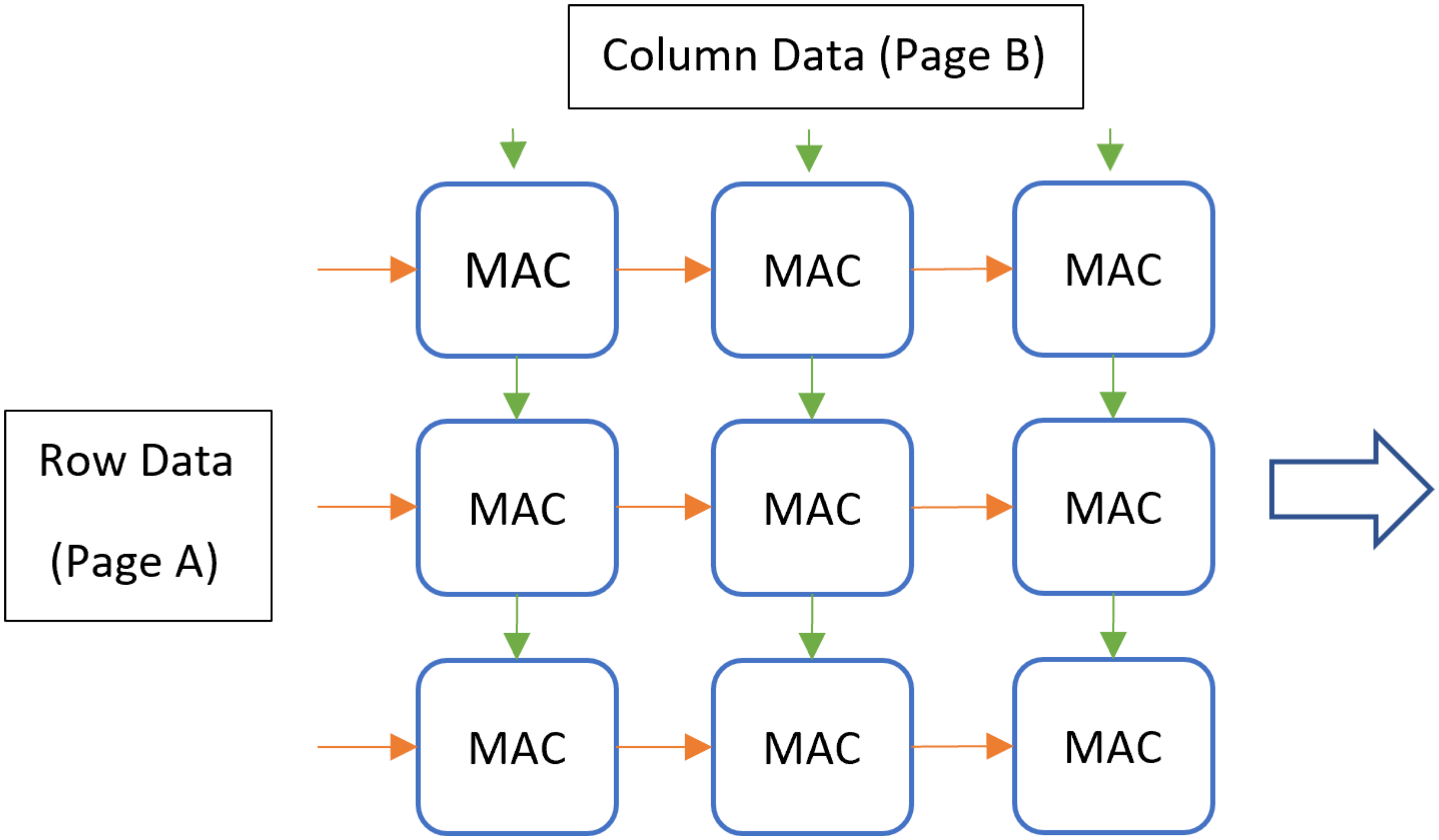
Ashish Tondwalkar

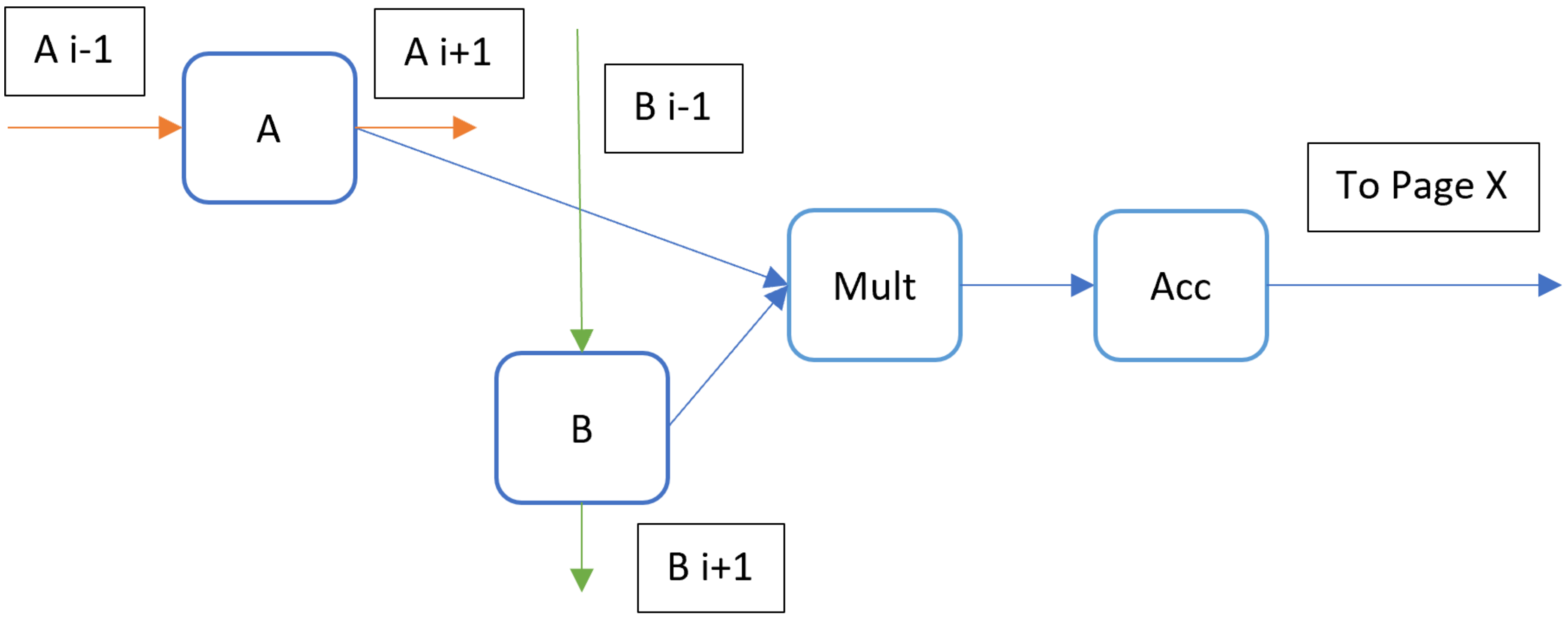

RISC
Control,
Data
DMA
Control
FIFOs
Data
Control
Data
Data
Cache
Data
DM
MXU

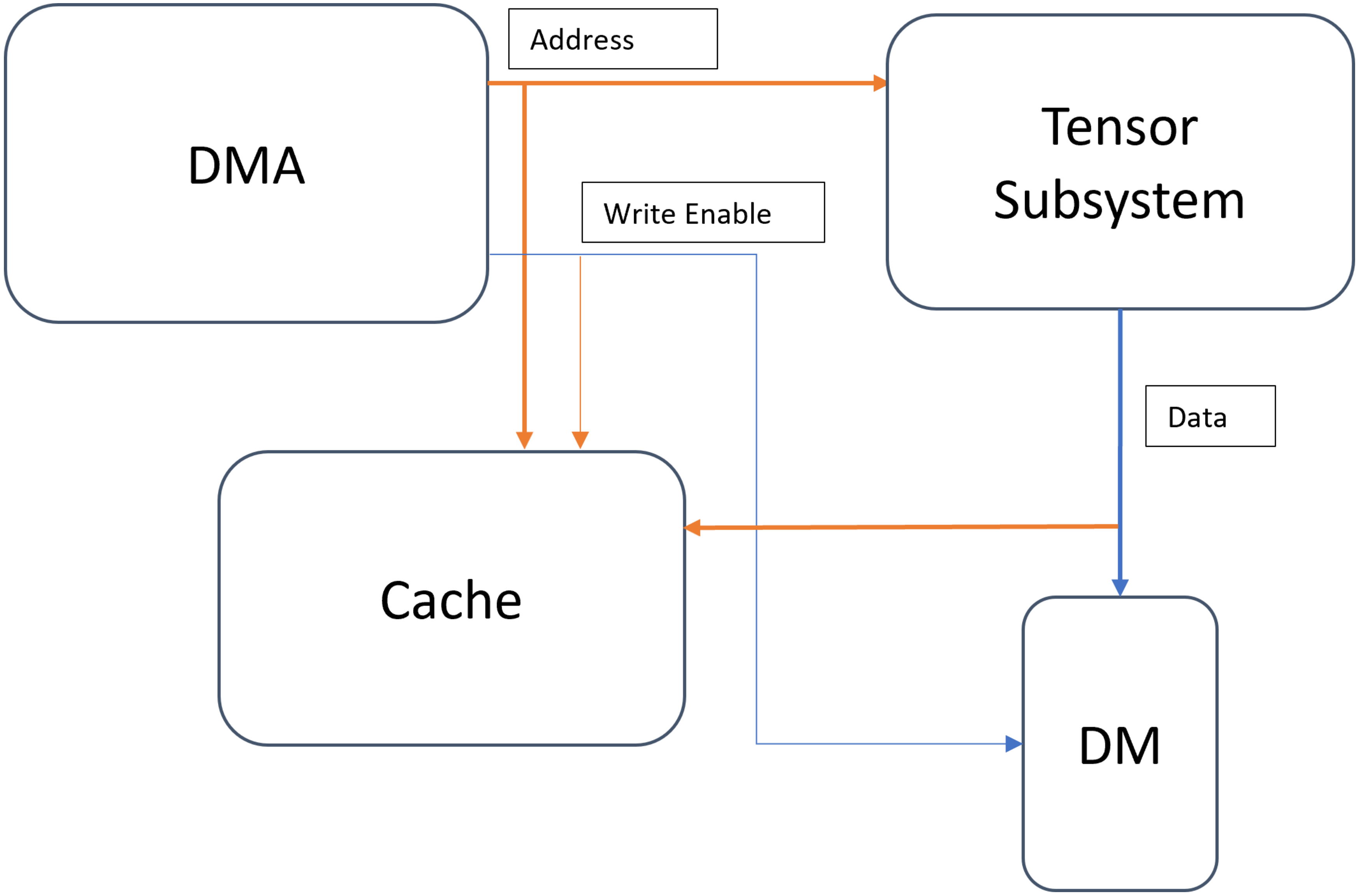


Data
RF Select
DMA
RF
FSM
Dimensions
FIFO Select
Address
Control Signals
DMA snoop

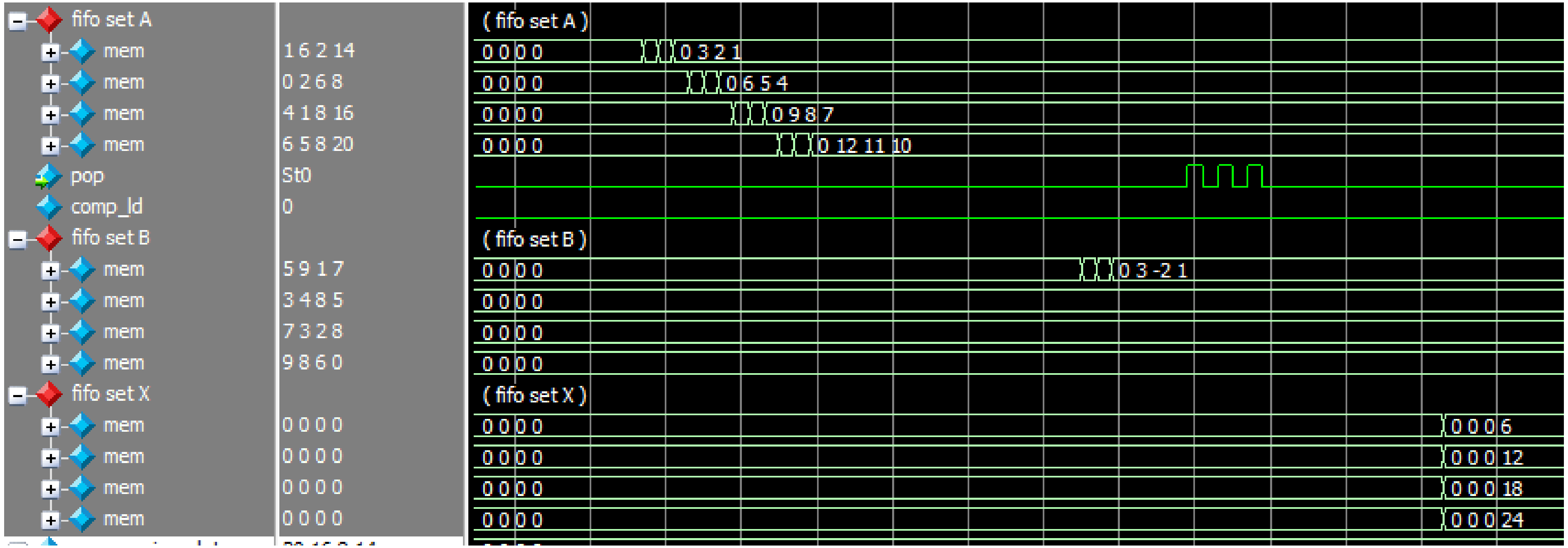

From DM
FIFO
Page B
Depth
Width
To DM
Depth
Width
FIFO
Page A
MXU DP
(Systolic
Array)
+
ReLU
FIFO
Page X
From DM
Weight
Mem


Column Data (Page B)
Row Data
(Page A)
MAC
MAC
MAC
MAC
MAC
MAC
MAC
MAC
MAC

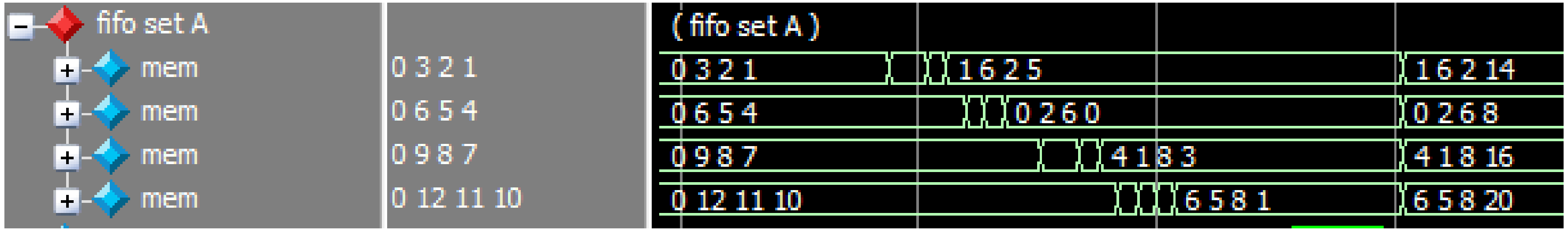

A i-1
A
A i+1
B i-1
B
B i+1
Mult
Acc
To Page X


DMA
Address
Tensor
Subsystem
Write Enable
Data
Cache
DM

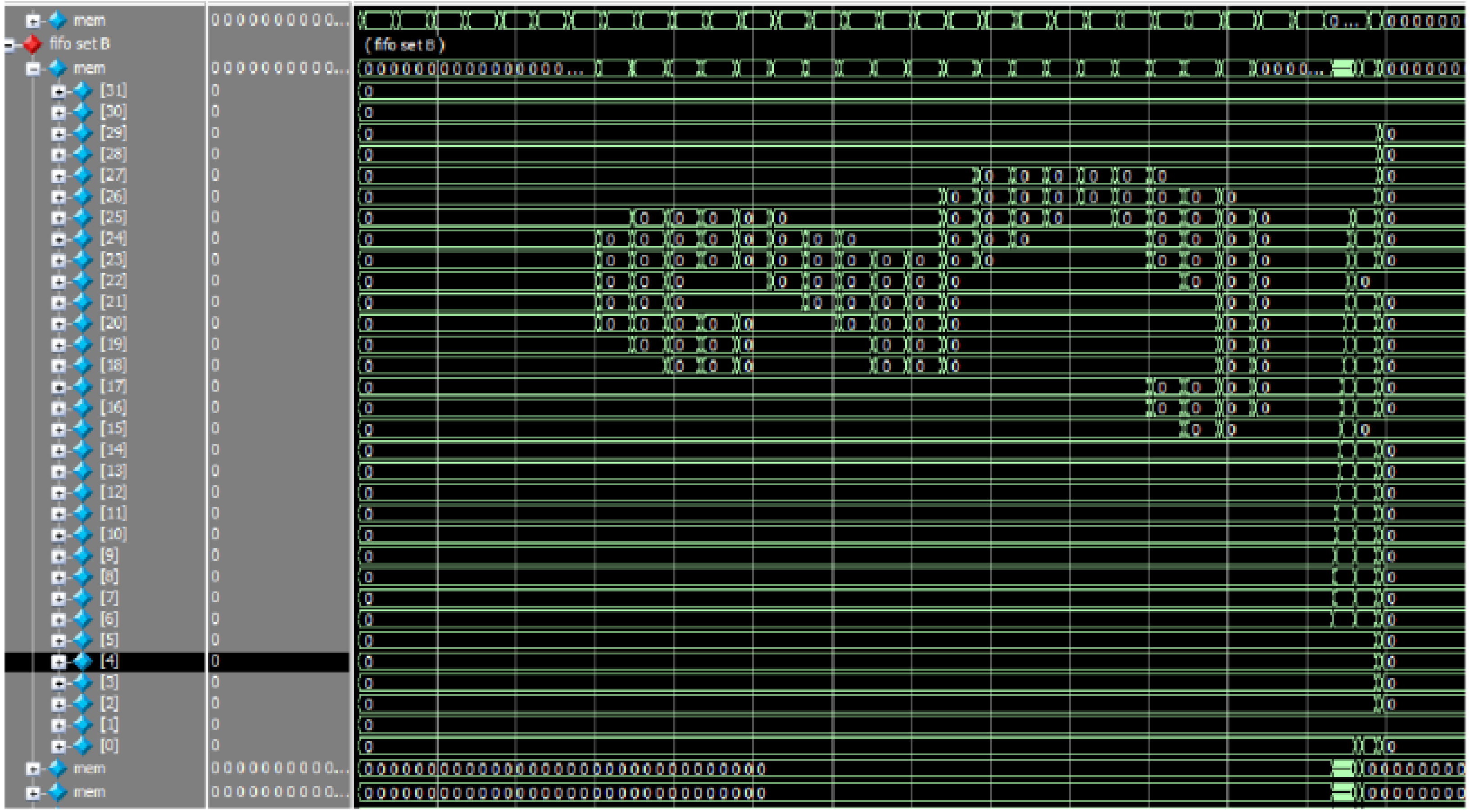
Matrix Multiplication



Column Compression



MNIST Network



Piecewise Multiplication

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000092c0	-14684	-26170	26215	29546	-23786	-15075	12790	11872	-9534	22350	-5421	-21918	-1333	-4623	24473	-17627
000092d0	11517	4942	22307	-18310	32121	28114	-1577	29805	-25363	7628	24264	32459	31389	6834	24155	435
000092e0	13943	26890	-25128	9304	27473	23559	-28107	15573	17998	-28904	-13807	-14580	27613	18606	-32031	1318
000092f0	31609	-25951	-11272	16927	-25233	6712	-7817	-23097	-6369	-30854	-4672	9500	-7552	-17174	14903	-7318
00009300	-9825	-16328	12753	1020	1399	19997	29398	-4105	12877	-2044	-24808	20929	32519	27332	-19223	5466
00009310	14127	522	-11601	30241	-5069	1080	-3894	-10261	12860	-4426	9463	13252	31563	1463	28783	-2259
00009320	-11566	-11047	-9174	-25382	-32673	-22888	-2910	30929	-24284	-32000	-27705	-8362	-10023	-13796	-25219	16225
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00009340	-32234	-2219	12349	15939	4857	12890	21363	-14506	18158	31544	12376	-30768	7248	8400	-6358	-6006
00009350	-20410	24600	-20972	21006	-10616	-31110	-6204	17214	28388	7814	-9924	22928	-20294	-5712	-3574	-1884
00009360	26624	-22212	1008	-10030	-25716	-30084	-16746	9194	17858	2585	-17976	-3044	-15293	25302	-31229	29272
00009370	1785	22048	24958	-30397	-6071	-10974	5484	-21717	12936	-9970	14368	27732	-21841	-20396	-25682	3739
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000093c0	-11274	-4980	-23808	-1091	-14375	-1932	-2212	-24991	-21512	29536	-22395	531	26188	30049	-12737	16357
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00009470	-32371	-5787	-10592	-16514	18978	-477	10178	18836	13642	25344	18166	22316	-18544	5108	-9194	-7648
00009480	-21598	32546	16166	7906	-13274	-11672	-2362	8784	19310	13270	12590	1850	-23414	-27830	21936	-8778
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000094b0	6067	-3803	31693	23229	20155	26260	-14234	17170	-2201	32412	29544	-32652	-25045	-4173	22960	-30002
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000094e0	2532	-16992	27851	-22122	7194	-26243	-23643	-23841	-17974	11657	23170	-28387	-4001	6054	5628	-18904
000094f0	-706	6398	-11771	11620	2027	-30562	4909	3341	32542	-13006	-1962	16970	-22546	-550	8857	938
00009500	27855	-11263	-24681	15739	-6170	-3878	28016	-18054	-16289	-15942	-17802	-6027	0	0	0	0
00009510	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Compression

0000f2b0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f2c0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f2d0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f2e0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f2f0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f300	379	0	4837	16047	11074	8125	2626	1661	5126	13378	14373	13131	14708	18255	12428	0
0000f310	15737	3439	7432	12440	10722	16083	0	11867	8023	12735	13344	14483	4680	10240	0	0
0000f320	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f330	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f340	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f350	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000f360	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

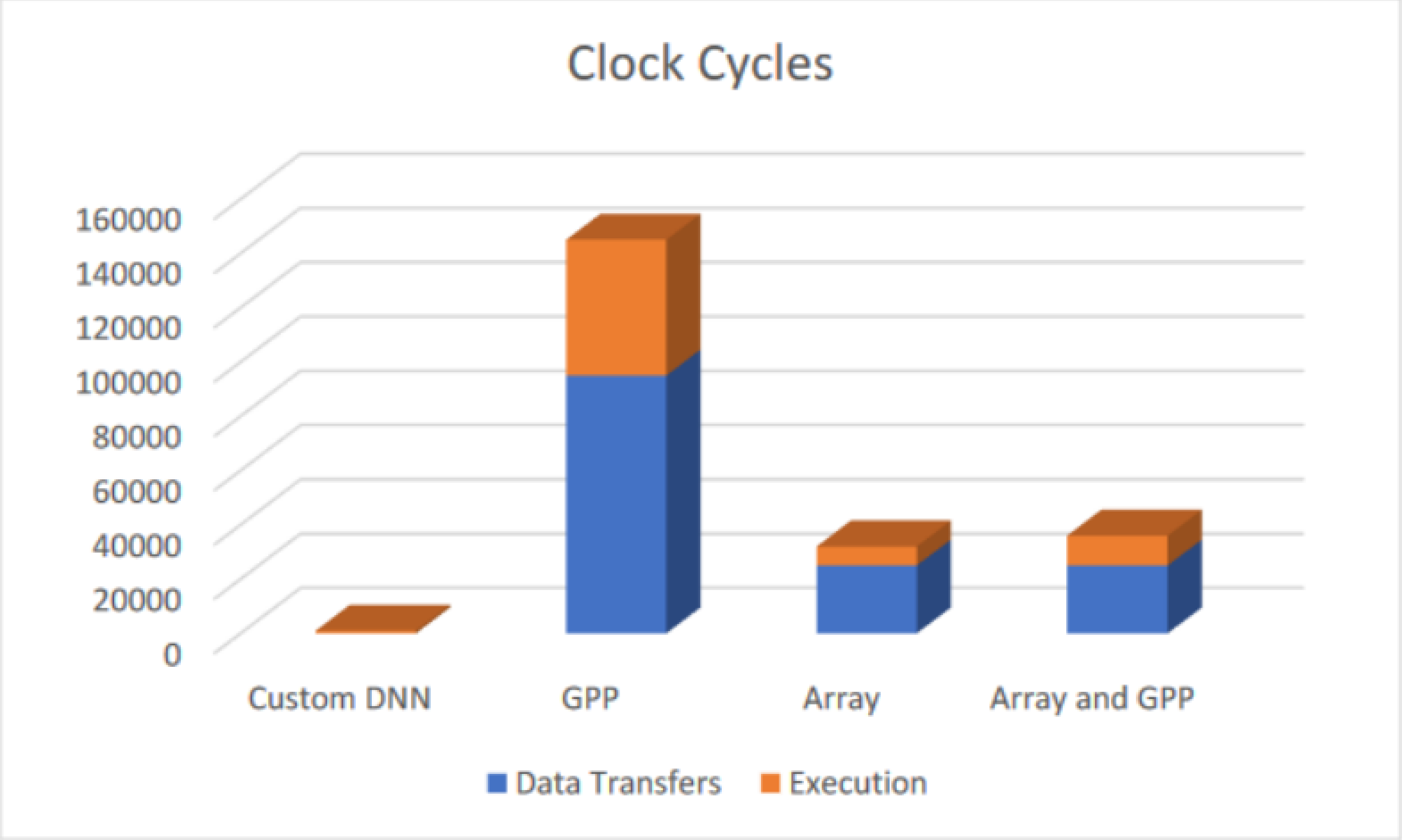
Output Layer

0000f4d0	0	0	0	0	0	0	0	0	0
0000f4e0	0	0	0	0	0	0	0	0	0
0000f4f0	0	0	0	0	0	0	0	0	0
0000f500	0	2236	10599	27767	0	12929	0	0	2817
0000f510	0	0	0	0	0	0	0	0	0
0000f520	0	0	0	0	0	0	0	0	0
0000f530	0	0	0	0	0	0	0	0	0

Metrics

	Full Program Clock Cycles	Data Transfer Clock Cycles	Execution Clock Cycles	LE	Regs	BM bits
Custom DNN	1000		1000	4066	2932	759296
GPP	145000	95000	50000	2015	681	32256
Array	32000	25000	7000	108340	126658	2101248
Array and GPP	36000	25000	11000	108340	126658	2101248

Execution



Clock Cycles
160000
140000
120000
100000
80000
60000
40000
20000
0
Custom DNN
GPP
Array
Array and GPP
Data Transfers
Execution

Utilization